STACKS, MODULI OF CURVES: LEMMA-CURVES-DIAGONAL-SEPARATED-FP

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 508-512 of sources/stacks-moduli-curves/pinned-original-source.tex; proof span is lines 514-633, the complete author proof (118 body lines).

Editorial note. Navigation stubs: 18 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 1 of queue_moduli-curves.json (pb=4687, stmt 508-512, proof 514-633).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 07XM: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DQ0: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 434–438. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DPP: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DPR: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DPN: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DPQ: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof

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but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9T: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D55: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D43: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D1I: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CMW: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CMY: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06Q3: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DNP: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04XD: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AYR: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AYX: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BEM: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 633–633. Cited by the retained statement or proof

but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The diagonal of Curves is separated and of finite presentation.*

Proof. Recall that *Curves* is a limit preserving algebraic stack, see Quot, Lemma 0D55. By Limits of Stacks, Lemma 0CMW this implies that $\Delta : \mathcal{P}olarized \to \mathcal{P}olarized \times \mathcal{P}olarized$ is limit preserving. Hence Δ is locally of finite presentation by Limits of Stacks, Proposition 0CMY.

Let us prove that Δ is separated. To see this, it suffices to show that given a scheme U and two objects $Y \to U$ and $X \to U$ of *Curves* over U, the algebraic space

$$Isom_U(Y, X)$$

is separated. This we have seen in Moduli Stacks, Lemmas 0DPN and 0DPP that the target is a separated algebraic space.

To finish the proof we show that Δ is quasi-compact. Since Δ is representable by algebraic spaces, it suffices to check the base change of Δ by a surjective smooth morphism $U \to \mathcal{C}urves \times \mathcal{C}urves$ is quasi-compact (see for example Properties of Stacks, Lemma 04XD). We choose $U = \coprod U_i$ to be a disjoint union of affine opens with a surjective smooth morphism

$$U \longrightarrow PolarizedCurves \times PolarizedCurves$$

Then $U \to \mathcal{C}urves \times \mathcal{C}urves$ will be surjective and smooth since $PolarizedCurves \to \mathcal{C}urves$ is surjective and smooth by Lemma 0DQ0. Since *PolarizedCurves* is limit preserving (by Artin's Axioms, Lemma 07XM and Quot, Lemmas 0D55, 0D43, and 0D1I), we see that $PolarizedCurves \to \mathrm{Spec}(\mathbf{Z})$ is locally of finite presentation, hence $U_i \to \mathrm{Spec}(\mathbf{Z})$ is locally of finite presentation (Limits of Stacks, Proposition 0CMY and Morphisms of Stacks, Lemmas 06Q3 and 0DNP). In particular, U_i is Noetherian affine. This reduces us to the case discussed in the next paragraph.

In this paragraph, given a Noetherian affine scheme U and two objects $(Y, \mathcal{N})$ and $(X, \mathcal{L})$ of *PolarizedCurves* over U, we show the algebraic space

$$Isom_U(Y, X)$$

is quasi-compact. Since the connected components of U are open and closed we may replace U by these. Thus we may and do assume U is connected. Let $u \in U$ be a point. Let Q, P be the Hilbert polynomials of these families, i.e.,

$$Q(n) = \chi(Y_u, \mathcal{N}_u^{\otimes n}) \quad \text{and} \quad P(n) = \chi(X_u, \mathcal{L}_u^{\otimes n})$$

see Varieties, Lemma 0BEM. Since U is connected and since the functions $u \mapsto \chi(Y_u, \mathcal{N}_u^{\otimes n})$ and $u \mapsto \chi(X_u, \mathcal{L}_u^{\otimes n})$ are locally constant (see Derived Categories of Schemes, Lemma 0B9T) we see that we get the same Hilbert polynomial in every point of U. Set

$$\mathcal{M} = \mathrm{pr}_1^*\mathcal{N} \otimes_{\mathcal{O}_{Y \times_U X}} \mathrm{pr}_2^*\mathcal{L}$$

on $Y \times_U X$. Given $(f, \varphi) \in Isom_U(Y, X)(T)$ for some scheme T over U then for every $t \in T$ we have

$$\begin{aligned} \chi(Y_t, (\mathrm{id} \times f)^*\mathcal{M}^{\otimes n}) &= \chi(Y_t, \mathcal{N}_t^{\otimes n} \otimes_{\mathcal{O}_{Y_t}} f_t^*\mathcal{L}_t^{\otimes n}) \\ &= n \deg(\mathcal{N}_t) + n \deg(f_t^*\mathcal{L}_t) + \chi(Y_t, \mathcal{O}_{Y_t}) \\ &= Q(n) + n \deg(\mathcal{L}_t) \\ &= Q(n) + P(n) - P(0) \end{aligned}$$

by Riemann-Roch for proper curves, more precisely by Varieties, Definition 0AYR and Lemma 0AYX and the fact that f_t is an isomorphism. Setting $P'(t) = Q(t) + P(t) - P(0)$ we find

$$Isom_U(Y, X) = Isom_U(Y, X) \cap Mor_U^{P', \mathcal{M}}(Y, X)$$

The intersection is an intersection of open subspaces of $Mor_U(Y, X)$, see Moduli Stacks, Lemma 0DPP and Remark 0DPQ. Now $Mor_U^{P', \mathcal{M}}(Y, X)$ is a Noetherian algebraic space as it is of finite presentation over U by Moduli Stacks, Lemma 0DPR. Thus the intersection is a Noetherian algebraic space too and the proof is finished. □